THERE IS SUCH THING AS A FREE LUNCH: SCHOOL MEALS, STIGMA, AND STUDENT DISCIPLINE

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Abstract

The Community Eligibility Provision (CEP) allows high-poverty schools to offer free meals to all students regardless of household income. Conceptualizing universal meal provision as a strategy to alleviate stigma associated with school meals, we hypothesize that CEP implementation reduces the incidence of suspensions, particularly for students from low-income backgrounds and minoritized students. We link educational records for students enrolled in Oregon public schools between 2010 and 2017 with administrative data describing their families' household income and social safety net program participation. Difference-in-differences analyses indicate that CEP has protective effects on the probability of suspension for students in participating schools, particularly for students from low-income families, students who received free or reduced-price meals prior to CEP implementation, and Hispanic students.

Keyword*:* Community Eligibility Provision, Free/reduced price lunch, Suspension, Universal meals

JEL Classification: I21, I24, I28, I32

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"I learned I was poor on the new lunch lady's first day at work."

- Psychologist Kenneth Payne, The Broken Ladder: How Inequality Affects the Way We Think, Live, and Die (2017, p. 11)

The National School Lunch Program (NSLP) provides free or subsidized meals to some 30 million students across the United States on the average school day, at a cost of approximately $14 billion annually (USDA 2022). One of the largest child nutrition and development programs in the contemporary U.S., NSLP feeds young people across the country who are at risk of food insecurity (Gordon et al. 2007).

However, NSLP is a means-tested program. To implement NSLP, schools collect data on students' family income and use these data to determine which students are eligible to receive free or subsidized meals. In doing so, schools enact social categories that are closely linked to student socioeconomic background. Daily rituals associated with the implementation of the NSLP such as lunch lines and meal tickets likely reinforce these income-based categories (Best 2017; Poppendieck 2011).

In this paper, we investigate the consequences of the Community Eligibility Provision (CEP) of NSLP. Authorized as part of the 2010 Healthy, Hunger-free Kids Act and rolled-out nationally beginning in the 2014-15 school year, CEP allows high poverty schools to offer free meals to all students regardless of their family's household income, suspending processes that distinguish between students who receive free or subsidized meals and those who must pay full price. As of the 2022-23 academic year, nearly 82 percent of eligible schools participated in CEP and nearly 20 million children attended a school that had adopted the program (FRAC 2023).

Our work conceptualizes CEP as an intervention that alleviates a source of socioeconomic stigma in participating schools. Prior research suggests that the implementation of CEP and other universal school meal programs boost student achievement (Gordanier et al.

2020; Kitchen et al. 2013; Ruffini 2021, Schwartz & Rothbart 2020) and attendance (Bartfeld, Berger, & Men 2020). Drawing upon prior scholarship that positions stigma at the heart of a vicious cycle that drives exclusionary discipline (Okonofua et al. 2016) – we focus attention on the effects of CEP on student suspensions. Consistent with the limited available evidence on the program's disciplinary consequences (Gordon & Ruffini 2020; Kho 2018), we hypothesize that CEP reduces the incidence of suspensions in treated schools. We further hypothesize that the program's benefits will be particularly pronounced for students from racially or ethnically minoritized backgrounds and students from socioeconomically disadvantaged families, since these students were most likely to encounter the stigma associated with means-tested lunch provision.

Our analyses take advantage of a unique dataset consisting of administrative data describing all students enrolled in Oregon public schools between 2010 and 2017 merged with household-level data from the U.S. Census Bureau describing Internal Revenue Service (IRS)-reported earnings and Supplemental Nutrition Assistance program (SNAP) participation. Since schools that implement CEP report free and reduced-price lunch enrollment data unevenly (Spiegel, Clark, Domina, Radsky, Yoo & Penner 20222), these merged data provide unprecedented opportunities to understand the extent to which the effects of CEP implementation vary with student socioeconomic background.

We use a difference-in-differences study design, which takes advantage of CEP's staggered implementation across eligible schools to estimate changes in the probability of suspension associated with CEP implementation for all students in participating schools. We further examine the way these changes vary with student household income and race/ethnicity. Assuming suspension trends for schools that implement CEP would move in parallel to trends

for schools that do not, if not for the implementation of CEP, this strategy generates unbiased treatment effect estimates. Although CEP rolled out in a period of rapid decline in the use of suspensions in Oregon schools, our analyses indicate that this identifying assumption holds.

CEP implementation corresponds with a further decline in suspensions for treated schools following the dramatic decline in suspension rates for Oregon's public schools generally in the years prior to CEP's launch (Hwang, et al. 2022). We estimate a statistically significant treatment effect equivalent to an approximately 10 percent decline in suspension probability generally, and nearly a 22 percent decline for students in grades 9-12. We find that CEP has a particularly pronounced protective effect for low-income students, as well as Hispanic students (the state's largest minoritized student group). Supplementary analyses estimated on a balanced panel of high school students indicate that the effect of CEP implementation was substantially larger for students who received free or reduced-price meals in the year before their school implemented CEP than for their peers who did not receive free or reduced-price meals prior to program implementation. While we are unable to directly measure lunch stigma and its consequences, these subgroup findings are consistent with the idea that CEP reduces suspensions by ameliorating the stigma associated with school meals in means-tested environments.

Community Eligibility Provision and universal school lunch

Meal programs in U.S. schools

Founded in 1946 and expanded considerably in the late 1960s and early 1970s, the NSLP targets free and low-price meals to students at risk of food insecurity (Levine 2008). Under program regulations, schools must provide NSLP applications to all students at the beginning of the school year and make applications available upon request at any time. These applications instruct potential applicants to report their household's total income from all sources in the most recent month. Students qualify for free lunches if their families report household incomes less

than 130 percent of poverty and reduced-price lunches if they report income between 130 and 185 percent of poverty. Students who do not apply for the program or whose families report income greater than 185 percent of poverty must pay for school lunches. Students who demonstrate income-eligibility remain enrolled for the entirety of a school year. Prior research indicates that enrolling in free or reduced-price lunch programs increases students' consumption of calcium, meat, and vegetables (Gordon et al. 2007) and a wide array of essential vitamins and minerals (Gleason & Suitor 2003), though school meals are associated with excessive sodium consumption (Clark & Fox 2009).

Several policy initiatives implemented since the 1970s have attempted to streamline the process of enrolling in free or reduced-price lunch. Students whose families participate in SNAP, Temporary Assistance for Needy Families (TANF), and other means-tested federal programs are automatically enrolled to receive free lunches without submitting an application (USDA 2008). Other policy initiatives allow participating schools to automatically re-enroll students who had previously demonstrated income eligibility without completing an annual application.

Authorized as part of the 2010 Healthy, Hunger-free Kids Act of 2010, CEP dramatically expanded and accelerated the policy movement to streamline access to free school meals. Schools, groups of schools within a district, or entire districts in which at least 40 percent of students are directly certified or whom school districts identify as homeless, migrant, or foster children can opt into CEP. Once they do so, participating schools offer free lunch and breakfast to all students, regardless of their demonstrated need, and receive reimbursement based on the percentage of their students in the aforementioned categories.[i] Since this formula reimburses eligible schools and districts with high concentrations of poor students at a higher rate than eligible schools and districts with lower concentrations of poor students, some participating

schools must access other revenue sources to cover universal breakfast and lunch expenses. As a result, CEP participation decisions vary both across schools and districts and within schools and districts over time.

Existing evidence on the effects of universal school meals

Recent evaluations of CEP implementation in U.S. schools as well as studies of universal meal provision in other policy contexts indicate that universal meal provision boosts student participation in school meals (Schwartz & Rothbart 2019; Ruffini 2021; Turner, Guthrie & Ralston 2019; Pokorney, Chandran, & Long 2019; Rothbart, Schwartz, & Gutierrez 2020; Gordanier et al. 2020; Kitchen et al. 2013) and student body mass index (Davis, Kreisman, & Masaddiq 2023). While it is difficult to directly compare the magnitude of changes in meal participation across settings, the results typically suggest that students in schools with universal lunch provision are 5-10 percentage points more likely to participate in school lunch than they were in means-tested lunch provision. This literature provides some evidence to suggest that increases in meal participation are particularly pronounced for the relatively economically advantaged students who would have previously had to pay for meals. However, these studies find positive effects in meal receipt for lower-income free or reduced-price eligible students as well.

This growing research literature further indicates that universal meal provision positively affects students' academic and behavioral outcomes. Several studies indicate that universal meal provision modestly but significantly boosts student math achievement, with average effect size estimates ranging from 0.04 to 0.06 standard deviations (Bartfeld, Berger, & Men 2020; Gordanier et al. 2020; Kitchen et al 2013; Ruffini 2021). While there is some evidence to suggest that universal meals have similar positive effects on reading achievement (Schwartz & Rothbart

2019), other studies show no effects on this achievement outcome (Gordanier et al. 2020; Ruffini 2021). Further, the existing literature indicates that universal meal provision improves elementary school student attendance (Gordanier et al. 2020; Bartfeld, Berger & Men 2020), though it is unclear whether these positive effects generalize to secondary school settings.

Relatively little evidence is currently available regarding the effects of universal meals on student disciplinary outcomes. Using data from New York City, Gutierrez (2020) demonstrates that school-level universal free meals program adoption drives notable improvements in student perceptions of school climate, including reported declines in bullying and fighting in schools. Gordon & Ruffini (2020) take advantage of the fact that eight states participated in a CEP pilot program starting in 2012 (before the program extended nationwide in 2015) to estimate the effects of attending a CEP-eligible school on suspension rates and other disciplinary outcomes using data from four waves of the Department of Education's Civil Rights Data collection. Their analyses suggest that suspension rates for elementary and middle school students in CEP-eligible schools fall modestly when CEP becomes available to schools in their states, though this decline is only statistically significantly different from zero for white males. While informative, these estimates likely understate the effect of school-level CEP participation, since only half of eligible schools had opted into the program by 2017. More directly relevant are findings from Kho's (2018) work using data from Tennessee. Using a difference-in-differences approach, Kho finds that student suspension rates remain unchanged during a school's first year of CEP participation but decline by 1.5 percentage points in their school's 2nd year in the program and 2.3 percentage points in their school's 3rd year in the program.

Lunch stigma and the effects of universal meals

The above-referenced evaluations of CEP and other universal meal programs typically invoke stigma as a mechanism for program effects. Link & Phelan (2001, p, 367) provide a

helpful definition of this widely-used term, writing that "we apply the term stigma when elements of labeling, stereotyping, separation, status loss, and discrimination co-occur in a power situation that allows the components of stigma to unfold."

Several studies suggest that the means-tested distribution of free or reduced-priced meals creates stigma by associating school meals with poverty. Students who come of age in schools that provide free or reduced-price meals to income-eligible students often use terms such as "disgusting" and "welfare food" to describe school meals (Poppendieck 2010). In these contexts, participation in means-tested free- or reduced-price school lunch programs may mark students as disadvantaged in the eyes of their teachers and peers (Weaver-Hightower 2011; Best 2017). This stigma may lead many students to opt out of school-provided meals (Mirtcheva & Powell 2009). Further, it likely raises the salience of student social class and other ascriptive student characteristics associated with social class in school status hierarchies, negatively affecting free lunch participants' achievement and other school-related outcomes (Yu, Lim, & Kelly 2019).

School meals and teacher perceptions

Generations of scholarship in the sociology of education draws attention to the ways student social class structures the relationship between teachers and administrators, who typically hail from middle class backgrounds, and their socioeconomically diverse students. Rist (1970) documents how information from social workers, fellow teachers, and interactions with parents about students' family backgrounds shapes teacher expectations of student behavior and ability even before they interact substantially with students. In schools in which school meal distribution is means-tested, teachers may distribute applications for free- or reduced-price lunch to students and their families, encourage families to apply, collect and submit applications, and organize students into lines at meal time. These activities may lead teachers to make inferences about

students' socioeconomic status that can shape their evaluations of students' academic ability (Ready & Wright 2011), motivation (Auwarter & Aruguete 2008; Jussim et al. 1987), and behavior (Dunkake & Schuchart 2015, Calarco 2018).

School meals and student social dynamics

Furthermore, while much of the school day is highly structured and adult supervised, school meal periods provide students with extensive and relatively independent social time. Student experiences during these periods influence both the way students see themselves and the way they are seen by others in the school environment (Milner 2013, Best 2017; Calarco 2018). As students sort into school lunch groupings, they often develop peer subcultures, distinguished by distinct norms of dress and self-presentation, behavior, and status ranking in the school's social hierarchy. Reflecting on the school lunchroom's role in his high school's social life, one student recounts: "I always felt that my level of coolness was being judged by the respective status of each individual I was with." (Milner 2013, p. 61).

Means-tested lunch distribution may increase the salience of social class at mealtime and throughout the school day. Students must line up for school meals and collect their food before they sit and eat. Students who opt out of school meal programs, meanwhile, can often proceed straight to their meals. Furthermore, as a growing number of schools begin to offer unsubsidized a la carte meal options, students who can afford to do so may pay cash for more desirable snack foods (Best 2017; Poppendieck 2011). These means-tested school meal distribution processes sort and spatially segregate students, influencing how students socialize in the lunchroom (Best 2017).

Free school meals and school discipline

By allowing participating schools to offer free meals to all students, CEP renders means-tested meal distribution mechanisms unnecessary, arguably mitigating meal stigma and its consequences for students. Doing so, we hypothesize, has downstream consequences for the disciplinary climate in participating schools. We imagine two mechanisms through which these consequences might emerge:

First, universal meal provision may improve student behavior by improving student nutrition. Prior research clearly demonstrates that CEP and other universal meal provision programs boost student participation in school meal programs (Schwartz & Rothbart 2019; Ruffini 2021; Turner, Guthrie & Ralston 2019; Pokorney, Chandran & Long 2019; Rothbart, Schwartz, & Gutierrez 2020; Gutierrez 2021; Gordanier et al. 2020; Kitchen et al. 2013). While part of this effect is likely attributable to the reduction in the cost associated with school meals for students who would have had to pay for school meals as well as a reduction in the administrative burden associated with free or reduced-price meal receipt, stigma reduction is likely also an important driver of increased meal take-up (Holford 2015; Gutierrez 2021; Schwartz & Rothbart 2019). A separate literature indicates that youth hunger and malnutrition are associated with externalizing behaviors (Howard 2011; Kimbro & Denney 2015) and nutritional supports protect at-risk youth from suspensions and other forms of school discipline (Gennetian et al. 2016).

Second, CEP may interrupt vicious cycles that contribute to high rates of exclusionary discipline, particularly for Black, Native American, and Hispanic youth. Since SES varies with race and since race is associated with poverty in society more broadly, school-based practices that increase poverty's salience in school social organizations and highlight the association between race and poverty can trigger broader racial stigmas. Such stigmas may lead educators to

scrutinize minoritized students' behaviors unsympathetically (Gilliam et al. 2016; Okonofua & Eberhardt 2015) and cause students of color to believe that school is not for them and behave accordingly (Borman et al 2021; Goyer et al. 2019; Kupchik, 2010). Together these responses may widen racialized disciplinary disparities.

Hypotheses

Consistent with prior research regarding the effects of universal meal provision on student outcomes, we hypothesize that:

Hypothesis 1: CEP implementation reduces the incidence of suspensions in treated schools.

Our account emphasizes the reduction of meal stigma as a primary mechanism for program effects. We acknowledge, however, that stigma is only one of several potential mechanisms that could account for CEP effects. The remaining hypotheses examine patterns that, at least indirectly, reveal the potential role that stigma plays in CEP effects.

If CEP changes student outcomes by reducing administrative burden in schools, we would expect the benefits of CEP to be shared across school populations. Similarly, if CEP changes outcomes by providing cost reductions in school meals for children from previously ineligible families, we would expect its effects to be most pronounced for these relatively affluent students. By contrast, if CEP affects outcomes by reducing the stigma associated with school meals, we would expect its effects to be most pronounced for children from low-income households. The means-tested distribution of free school meals likely forced these students to either face stigma associated with receiving a free meal or avoid the stigma by foregoing school meal (and its nutritional benefit). In either case, our focus on stigma as a mechanism for CEP effects leads us to hypothesize that:

Hypothesis 2: CEP has a pronounced protective effect on suspensions for low-income students, and especially those who had previously been eligible to receive free or reduced-price meals.

Previous research provides evidence consistent with this second hypothesis (Mitcherva & Powell 2009, Kitchen et al. 2010, Gordanier et al. 2020, Bartfeld et al. 2020). However, this study's unique link between school and tax data makes it the first, to our knowledge, to directly investigate the ways the effects of universal school meal provision vary with student economic background.

Finally, in light of existing scholarship on disciplinary disparities focusing extensive attention on pronounced and persistent racial disparities in exposure to school discipline (Anderson & Ritter 2017; Barrett, McEachin, Mills & Valant 2021; Gregory, Skiba, & Noguera 2010; Lindsay & Hart 2017; Owens & McLanahan 2020; Freidus 2020), we might expect CEP's stigma-reducing effects to be most pronounced for racially minoritized students. If CEP implementation reduces the salience of these racial stereotypes, we hypothesize that:

Hypothesis 3: CEP has a pronounced protective effect on suspensions for racially and ethnically minoritized students. In the Oregon context, this effect may be most observable among Hispanic students, the largest minoritized racial/ethnic group among Oregon public school students.

Existing research on this point is limited. Gordon & Ruffini (2021) find evidence to suggest that the disciplinary consequences of attending a CEP eligible school are most pronounced for white male elementary-aged students, though, as noted above this study is based on CEP eligibility, not participation.

Data

To test these hypotheses, we link student-level educational administrative data from the Oregon Department of Education (ODE) with confidential microdata stored at the Census Bureau including SNAP records from the state of Oregon and income tax records from the IRS. The ODE data provide annual records on all students enrolled in Oregon public K-12 schools between 2009-10 and 2016-17. Included in these records is an indicator flagging each academic year in which a student has received either an in-school or out-of-school suspension. This indicator is the focal outcome for our paper. Additional measures include indicators of student race/ethnicity and annual indicators flagging students who were enrolled in free or reduced-price lunch enrollment for each year 2010-2014.[ii] Students appear in SNAP records when they are listed as members of households that receive SNAP benefits; they appear in IRS records when they are claimed as dependents on IRS tax returns.

Merging across these data sources is possible using Census's Protected Identification Keys (PIKs), which are assigned to the three administrative data sources using the Person Identification Validation System.[iii] As noted in Table 1, 96 percent of K-12 students in the ODE data are assigned PIKs in this process. We exclude students without PIKs from our analyses. Hispanic children and children in non-married households are overrepresented among these excluded students. We use inverse probability weights to address these non-response biases.[iv]

[TABLE 1 AROUND HERE]

Appendix A provides details on the variables we construct using the linked data. One variable of particular importance is the low-income indicator, which leverages both SNAP and IRS data to identify students who likely live in households with income below 1.85 times the federal poverty line (and are, thus, eligible for FRL). In addition to restricting our sample to

students who have PIKs, we further require a non-missing value of the low-income variable, since this variable is central to our heterogeneity analyses. Six percent of Oregon students do not have a low-income value, either because they do not have a PIK or because they do not appear in the IRS Form 1040 or SNAP records in the relevant years. By dropping students with missing data, we proceed with the implicit assumption that they are missing at random, which is supported by related research.[v] After restricting based on the availability of PIK and income information, the data is reduced to 4.2 million student-year observations.

To create our analytic sample, we additionally exclude students enrolled in schools that are not eligible to participate in CEP, as well as a small number of students enrolled in schools with fewer than 20 students, specialty schools managed directly by their district or the Oregon Department of Education, and schools that either opened or closed at any point during the study period. Schools are eligible for CEP if 40 percent or more of enrolled students directly certified for free school meals or identified by the school district as homeless, foster, or runaway youth in the prior academic year. Data on this measure, which the USDA refers to as the Identified Student Percentage (ISP), are available for Oregon between 2014-15 and 2016-17 from the Food Research and Action Center (FRAC) and the Center on Budget and Policy Priorities. However, since these data may be incomplete, we additionally use data on school SNAP rates, because SNAP participation is the main characteristic used to calculate ISP. Schools are marked as eligible for CEP if SNAP percentages are greater than 40 percent in the prior school year. Finally, because districts can group schools into CEP-eligible clusters based on the group ISP rate, we consider all schools in districts where the entire district has a SNAP percentage of at least 40 percent to be CEP eligible, regardless of the concentration of SNAP participating

students in the school. These sample restrictions exclude roughly one-half of the full data, for an analytic sample of 567,000 students and 2,104,000 student-year observations.

Table 1 highlights the ways that students in CEP eligible schools differ from the average Oregon student. Our analytic sample includes a higher proportion of low-income students, English language learners, and Hispanic students than the state overall. The analytic sample also has somewhat higher average suspension rates compared to excluded schools. Elementary schools are disproportionately likely to be eligible for CEP and, accordingly, CEP schools have lower enrollment than the state average.

Approximately one-in-five students in CEP-eligible schools are exposed to NSLP's Provision 2 in our data. Provision 2 provides free meals to all students, though less than 7 percent of Oregon schools provided universal free lunch via Provision 2 in 2014. Most participating schools opt to only provide free breakfast. Unlike CEP, Provision 2 requires use of the traditional school lunch application process to establish initial reimbursement rates, and the USDA does not provide additional funds to support school meals for students that are not income eligible (USDA 2002). Thus, the program is most financially viable for extremely high poverty schools. Perhaps reflecting the lack of financial incentive, many schools drop into and out of Provision 2 year over year. We view Provision 2 as a substantially weaker treatment than CEP, and one concentrated in extremely high poverty schools where free lunch is the norm and lunch stigma may be less salient. Our analyses do not differentiate between the effects of CEP implementation for schools that had previously participated in Provision 2 and schools that had not. In supplementary analyses, available by request, we exclude schools who participated in the Provision 2 from the analytic sample. These analyses return substantively similar results to the analyses reported here.

Empirical methodology

We employ a difference-in-differences (DiD) approach to estimate the effects of CEP implementation on suspension probabilities for students in CEP-eligible schools, with particular attention to heterogeneous treatment effects across the student subgroups most likely to confront stigma. We discuss our primary model below. Appendix B provides details on supplementary analyses that test for treatment effect variation with the timing of program implementation and the number of years schools have implemented the program.

Because the use of suspension is more common in middle and high school compared to elementary school, we estimate models separately for grades K-5, 6-8, and 9-12, by splitting the sample based on student grade, in addition to estimating treatment effects for the full sample. We further estimate treatment effects for two panels of students – those in grades K-2 in 2013-14, and those in grade 9 in 2013-14 – the details of which we describe in the student fixed effects section below.

Finally, we modify the models to allow treatment effects to vary by student characteristics, including low-income status and race. Thus, we offer estimates of the overall effect of CEP on suspension, broadly, as well as estimates by school level, student subgroups, and subgroups by school level.

Estimating the effect of CEP implementation

First, we model the treatment-on-the-treated effect for CEP overall. That is, if we assume the treatment effect of CEP is constant across cohorts and treatment periods, we can identify the treatment effect of CEP on student outcomes with the following model:

$$(1) \quad y_{itj} = \eta + \sum_{s=2011}^{2017} \theta_s fs_t + \tau TREAT_{tj} + \gamma_j + \epsilon_{it}$$

where $TREAT_{tj}$ is equal to 1 if school j in year t (in which student i is enrolled) is in CEP, and equal to 0 otherwise. We focus on a single outcome, so y_{itj} is an indicator for whether student i enrolled in school j is ever suspended in year t, and, since we estimate the above equation using OLS, we are estimating a Linear Probability Model. Treatment effects are identified under our estimating assumptions because school fixed effects (γ_j) remove level differences in suspensions rates across schools, and academic year indicators fs_t adjust for time trends in suspension probabilities common to all CEP-eligible schools.[vi] Standard errors for this model and all subsequent models are adjusted to reflect clustering at the district level. In addition to estimating Model (1) for the entire population of students, we estimate it separately on samples restricted to include only students in elementary, middle, and high school grades. We do not include additional covariates due to ongoing concerns in the literature about the interpretability and reliability of DiD-derived treatment effects in the presence of time-varying covariates (Caetano et al. 2022).

Implicit in Model (1) is an assumption that treatment effects are constant across cohorts and treatment periods (Goodman-Bacon 2021; Wooldridge 2021). We need not make this assumption. Thus, we estimate versions of Model (1) in which we break out the treatment indicator into three treatment period indicators, and, in its most disaggregated form, by treatment cohort-by-period indicators.[vii] Both models provide insight into the suitability of assuming a single constant treatment effect in this setting. The details of these disaggregated models are in Appendix B.

Three key assumptions must hold for Model (1) to generate unbiased causal treatment effect estimates: (1) schools did not adjust policies and practices in advance of program implementation, (2) in the absence of CEP implementation, outcomes in treatment schools would

have followed parallel trends to outcomes in untreated schools, and (3) students did not select non-randomly into schools on the basis of their CEP participation.

The first assumption is reasonable in this setting since eligible schools and districts act in late spring to enroll in CEP for the subsequent school year. It is unlikely that students, teachers, or administrators change their behavior with respect to suspension because they anticipate universal free lunch provision the next year. It is also unlikely that schools change their lunch practices prior to their official enrollment in CEP, since their reimbursements do not change until after enrollment.

The second assumption is somewhat more restrictive. The CEP program rolled out during a period in which educators, policymakers, scholars, and activists were directing considerable attention to school discipline practice and the ways these practices contribute to educational inequalities. As we discuss in more detail below, this scrutiny appears to have contributed to broad-based declines in school suspension rates. These trends do not in and of themselves constitute a threat to the unbiased estimation of the effects of CEP implementation using a difference-in-differences approach. However, if declines in CEP implementation correlate with school decisions to implement CEP, they may confound our treatment effect estimates. To address this potential threat to valid inference, we provide a detailed comparison of pre-treatment suspension trends across participating and non-participating CEP eligible schools, including estimating interaction analyses that test for significant variation in overall and group-specific suspension trends by treatment cohort.[viii]

The third assumption is also somewhat restrictive. When schools opt into CEP, they position themselves to provide material benefits for students and their families. Students who would not have been income-qualified for free meals benefit by receiving free lunches; students

who would have been income-qualified for free meals benefit from a reduction of administrative and social barriers around the take-up of these free meals. These benefits may attract new students to CEP-participating schools and/or increase the attachment of existing students to the school. To the extent these forces change the composition of the student body to include more students with disproportionately low suspensions rates, non-random selection into CEP-participating schools could modestly bias difference-in-differences estimates of the effects of CEP implementation in favor of finding a suspension-reducing effect. To address this possibility, we estimate a supplementary version of Model (1) with student fixed effects.

The student fixed effects model

In a variation on Model (1), we employ a panel dataset in which we observe students for four consecutive years in the same school. This offers two advantages: first, we can use student fixed effects to estimate the within-student effect of CEP; and, second, we can define student subgroups by their pre-CEP income and lunch status, to examine how pre-treatment labels may factor into the CEP treatment effect.

We estimate:

$$(2) \quad y_{itj} = \eta + \sum_{s=2015}^{2017} \theta_s fs_t + \tau TREAT_{tj} + \alpha_i + \epsilon_{it}$$

where α_i denotes student fixed effects. The other change relative to Model (1) is that we restrict attention to school years 2014-2017, which, given the perfectly balanced student panel we construct, means we observe all students in pre-treatment 2014, and for each of the next three years, as CEP is phased in.

This approach addresses the threat of non-random selection into CEP schools by eliminating students who move schools and focusing analytic attention on within-student changes in suspension outcomes associated with CEP implementation. However, the sample

restrictions required to arrive at a perfectly-balanced panel yields a non-representative sample of ODE students. In addition to excluding students who switch schools, we also exclude students missing a low-income value in any of the four years in order to maintain a balanced sample while matching the low-income exclusion of the main model. The grade spans we can examine are also limited: this sample includes only students who are in Kindergarten, first, second, or ninth grade in 2014, so we can observe them in the same school for three additional years. As such, we estimate this student fixed effect model separately for the elementary (grades K-2) panel and the ninth grade panel.

Estimating variation in effects of CEP across subgroups

Our second and third hypotheses predict differing effects of CEP for different subgroups of students. To test these hypotheses, we estimate an elaborated version of Model (1), in which treatment and academic year indicators are additionally interacted with subgroup indicators. For simplicity, consider subdividing students into two groups according to binary indicator B_i, which could denote having a low-income family or having a racially or ethnically minoritized identity.

$$(3) \quad y_{itj} = \eta + B_i + \sum_{s=11}^{17} \theta_s fs_t + \sum_{s=11}^{17} \pi_s fs_t B_i$$

$$+\tau TREAT_{tj} + \rho TREAT_{tj} B_i + \gamma_j + \epsilon_{it}$$

The difference in the treatment effect of CEP for students in group B compared to students not in group B is ρ. The treatment effect for group B students is $\tau + \rho$, while the treatment effect for students not in group B is simply τ.

We test our second hypothesis by estimating a version of Model (3) that compares the effects of CEP implementation on students from low-income households and students from households that are not low income. We test our third hypothesis by estimating a version of Model (3) that compares the effects of CEP implementation on American Indian/Alaska Native,

Asian/Pacific Islander, Black, Hispanic, Multiracial, and White students. Because we are interested in how the net treatment effects vary across subgroups, rather than present lengthy regression results, we estimate linear predictions of treatment effects for different subgroups. For example, when we estimate the regression where B_i is an indicator for low-income status, we report the treatment effect for students who are not low income ($\hat{\rho}$), and contrast it to the treatment effect for students who are low income ($\hat{\rho} + \hat{\tau}$).[ix]

We run the same analyses using the student fixed effects model, with analogous modifications to Model (2). The panel data in the student fixed effect model also allow us to test a more nuanced version of our second hypothesis, by comparing treatment effects for subgroups defined by their FRL enrollment and low-income status in 2014, prior to the implementation of CEP. This is important because prior research demonstrates that many income-eligible students do not enroll in means-tested meals programs while many seemingly income-ineligible students do, making the low-income measure an imperfect proxy of FRL enrollment, and thus an imperfect indicator of which students might have experienced FRL-related stigma (Domina, Pharris-Ciurej, Penner, Penner, Brummet, Porter, & Sanabria 2018). Moreover, means-tested meal enrollment status can only be reliably observed prior to CEP implementation (Spiegel et al. 2022). We therefore use the student fixed effects model to test CEP's disciplinary effects separately for students in four categories based on their 2014 statuses: (1) students who are not enrolled in FRL and are not income eligible, (2) students who are not enrolled in FRL but are income eligible, (3) students who are enrolled in FRL but not income eligible, and (4) students who are enrolled in FRL and income eligible. Like all of the student fixed effect estimates, this approach sacrifices external validity, but these estimates provide a unique opportunity to separate CEP's stigma-reduction mechanisms from its nutrition-provision mechanisms. We would expect

a nutrition pathway to particularly benefit low-income students not previously enrolled—who may be less likely to bring or purchase a nutritious lunch—while a stigma pathway would benefit students previously enrolled, regardless of their income status.

Results

Patterns in CEP implementation

Figure 1 maps schools that implemented CEP across Oregon from 2014-15 to 2016-17. This figure represents CEP participating schools as black diamonds and non-participating schools as white circles. Two hundred forty-two of the 316 Oregon public schools that took up CEP during our student period did so in 2015, 63 joined in 2016, and 11 joined in 2017. As the figure illustrates, CEP schools are distributed widely across the state of Oregon. The second map included in Figure 1 represents CEP participation in the Greater Portland region. Most larger school districts in Oregon such as the Portland and Salem-Keizer school districts include both CEP-participating and non-participating schools. Approximately 40 school districts throughout Oregon include a mixture of participating and non-participating schools. In 22 districts, all schools participated in CEP.

[FIGURE 1 AROUND HERE]

As noted in Table 1, CEP-eligible Oregon public schools enroll more low-income students and more students of color than CEP-ineligible schools since federal policy only allows schools and districts with high proportions of economically disadvantaged students to participate in CEP. This pattern, however, also extends to program take-up among eligible schools, as shown in Table 2, which compares the demographic profile of schools that implemented CEP during our study period (our treatment schools) with that of schools that were eligible to enroll in the program, but did not do so (our control schools). On average, treatment schools enroll a

higher proportion of low-income students and students of color than control schools. For example, in 2014, treatment schools are 51 percent White and 34 percent Hispanic compared to control schools which are 66 percent White and 21 percent Hispanic. 49 percent of students in treatment schools are enrolled in SNAP in March 2014 compared to 37 percent of students in control schools. The mean family income for students in treatment schools is $10,000 lower than for students in control schools. CEP's funding formula, which reimburses participating schools based on the proportion of students who participate in SNAP and other means-tested social safety net programs, likely accounts for these patterns.

[TABLE 2 AROUND HERE]

Patterns of change in treatment and control schools: Testing the parallel trends assumption

While our difference-in-differences approach effectively accounts for time-invariant differences between the treatment and control group, causal inferences based on this design rely on the assumption that trends between treatment and control groups would move in parallel if not for the treatment. While the parallel trends assumption is ultimately untestable, the descriptive data reported in in Table 2 is consistent with parallel trends. The data reported here reveal little divergence in covariate trends between treatment and control groups, suggesting that these groups of schools were exposed to similar pressures during the pre-treatment period and that – by extension – divergences in outcomes that emerge during the post-treatment period are likely attributable to the effects of CEP enrollment. The percentage of Hispanic students rises by four percentage points in both treated and untreated schools over the 4-year pre-treatment time frame. Schools in both groups also see small parallel increases in the proportion of students who speak a language other than English in the home between 2010 and 2014. Similarly, although the mean IRS-reported income for students in both sets of schools increases by approximately 5 percent

between 2010-2014, the percent of students enrolled in SNAP increases by four percentage points in each group during this pre-treatment period.[x]

Figure 2 reports trends in suspension rates, our outcome variable. These unconditional trends provide additional evidence to support the parallel trend assumption. In 2010, the percentage of students ever suspended was 1.3 percentage points higher for students attending treatment schools compared to students attending schools that never participated in CEP. While suspension rates remain largely unchanged between 2010 and 2011 for schools in the control group, we observe substantial declines in suspension rates for both treated and control groups between 2011 and 2014. This rate of decline corresponds to a broad national trend in school discipline (Hirschfield 2018). As Hwang et al. (2021) point out, these declines largely preceded the passage of two laws passed by the Oregon state legislature in 2013 and 2014 designed to eliminate zero tolerance school disciplinary policies and reduce the incidence of school suspensions for relatively minor infractions. In 2015, however, the first year of CEP, suspension rate trends for students in treatment and control schools begin to diverge. Students in schools that implemented CEP see a sustained decline in suspension rates, while the trend in suspensions among students in control schools levels off and even ticks upward in 2017. By 2017, despite having higher rates of poverty and attending schools with higher concentrations of poverty, students in schools that implemented CEP have lower rates of suspension compared to students in schools that declined to participate in CEP.

[FIGURE 2 AROUND HERE]

Table 3 reports the results of a more formal test of the parallel pre-trends assumption. We conduct this test by running a single regression of suspension on indicators for academic year, treatment cohort, and the interaction of the two, omitting academic year 2014 as the refence

category. The first column of coefficients in Table 3 represents the difference in suspension probabilities for each year compared to 2014 for students in control group schools. Consistent with the results reported in Figure 2, this column indicates that suspension probabilities declined by approximately two percentage points between 2010 and 2014 but remained largely unchanged from 2014-17 for students in control schools. The coefficients reported in the remaining columns show the deviation from the control group differences for each treatment cohort. Specifically, the coefficients in bold offer our test of pre-trends. These coefficients are consistently small in magnitude and not statistically significant. As such, they provide no evidence of systematic divergence in suspension outcomes between treatment and control groups in the periods prior to treatment.[xi] We see in Table 3 strong evidence in support of the parallel trends assumption that underlies the difference-in-differences analyses that we present below.

[TABLE 3 ABOUT HERE]

In Appendix Table 1, we replicate these analyses for subsamples of students in elementary, middle, and high school grades. While these subgroup-specific pre-trend analyses are somewhat less precise than the main analyses, they return largely similar null results. We additionally test the parallel pre-trend assumption using a series of linear pre-trend tests conducted on a sample that is limited to pre-treatment academic years for each given cohort, These analyses, which are available by request, suggest suspension trends in treated schools did not significantly diverge from suspension trends in control schools, and pre-trends are parallel among low-income and racially minoritized students.

Estimating the effects of CEP implementation

We next present the results of difference-in-differences models that assess the average effects of CEP implementation and estimate the extent to which these effects are sustained or

grow in the years after treatment. Table 4 summarizes two sets of effects for the full sample: those from the three-period model, which allows treatment effects to vary across treatment period, and those from the Model (1), which assumes a common treatment effect across cohorts for each period of treatment. Appendix Table 2 reports the more detailed cohort-by-period results, as well as the three-period results for all subsamples.

[TABLE 4 AROUND HERE]

In the period-specific model, only the third treatment period yields a statistically significant negative effect for the full sample. This negative coefficient suggests that in its 3rd year of implementation CEP reduced suspension probabilities by approximately one percentage point (from a baseline of approximately 7.3 percent). The coefficients for the first two treatment periods are negative, but not quite statistically significant. However, we cannot reject the null hypothesis that treatment effects are the same in each period (as reflected in the p-value at the bottom of three-period model results in Column 1). In Appendix Table 2, we replicate this finding for grade-level specific subsamples. The constant treatment effect hypothesis is particularly compelling in grades 9-12, which experience the largest overall treatment effects from CEP. Based on this finding, we report single CEP treatment effects, as described in Model (1) in all subsequent analyses. Column 2 of Table 4 reports this single treatment effects estimate for the full sample, which indicates that the average effect of CEP implementation is a 0.7 percentage point reduction in student suspension probabilities.

The analyses reported in Columns 3, 4, and 5 of Table 4 consider CEP's disciplinary effects separately for students in elementary, middle, and high school grades. We find no significant effect of CEP implementation on suspension in the elementary school grades, but substantial and statistically significant protective effects on suspension in middle and high school

grades. We find that CEP implementation reduces the probability of suspension for students in middle school grades by nearly 2 percentage points (a 12 percent reduction from the pre-CEP mean). For students in high school grades, CEP implementation reduces the probability of suspension by nearly 2.5 percentage points (a 22 percent reduction from the pre-CEP mean.)

Although the student fixed effect models are estimated on sharply reduced sample of students who remain in the data and in the same school from 2014 through 2017, the results of these models, reported in columns 6 and 7 of Table 4, are largely consistent with our primary school fixed effect model findings. We find no evidence of a CEP effect on suspension in the elementary student fixed effect model, in which students were generally first exposed to CEP in grades 1-3. However, the 9th grade student fixed effect model, in which most students experienced CEP starting in 10th grade, indicates that CEP implementation reduces student suspension probabilities by nearly 3 percentage points. This is a very large effect given this selected sample has a lower baseline suspension rate than students in grades 9-12 generally. Not only is the prevalence of suspension much higher in higher grades, but we suspect the experience of stigma is also more pervasive in higher grades.

In summary, consistent with our first hypothesis, we find that CEP implementation reduces suspensions in participating schools, though effects in elementary schools are not statistically significant.

Evidence of effect heterogeneity

Our second and third hypotheses suggest that reductions in suspensions following the introduction of universal free lunch should be concentrated among students most likely to experience lunch-related stigma: low-income students and students of color. A unique contribution of our paper comes in our ability to test these hypotheses by analyzing heterogeneity

in CEP treatment effects across socioeconomic and racial subgroups. To do so, we estimate Model (3) using linear probability models with school or student fixed effects, and then report linear predictions of subgroup-specific treatment effects in Table 5. These estimates can be interpreted as the treatment effect on suspension of CEP for students in participating schools depending on their subgroup membership.

[TABLE 5 AROUND HERE]

The analysis reported in Panel A of Table 5 focuses on the extent to which the effects of CEP vary with student economic background. We find that low-income students[xii] experience large and statistically significant treatment effects. The provision of universal school meals under CEP reduces suspension probabilities for low-income students by nearly a percentage point. This percentage point difference appears for high school students, too, while middle school students are unique in having a similarly sized and statistically significant effect for both low-income and not-low-income students.

In Panel B, we contrast three levels of income, corresponding approximately to reduced-price and free lunch eligibility cut-offs. We find free lunch eligible students experience the largest effects overall and in high school, but reduced-price lunch eligible students also experience significant suspension declines. In elementary and middle school, there is no distinction between reduced-price and free lunch groupings.

In Panel C, we consider the extent to which the effects of CEP vary with student race and ethnicity. These estimates show that Hispanic students experience considerably larger treatment effects than students in all other racial/ethnic groups. These models suggest that CEP implementation reduces Hispanic students' suspension probabilities by 1.4 percentage points across the entire sample and by nearly 3 percentage points for students in middle school grades,

with a similar effect size in high school that is not quite statistically significant. While the effect estimates for White students in these models are also negative, they are much smaller than the estimated effects for Hispanic students and are not statistically significant. Effects for other racial subgroups are ambiguous – likely due to their small representation in Oregon public schools. Black students show statistically significant drops in suspension probabilities similar to Hispanic students in elementary and high school, though the adverse and imprecise effect in middle school is difficult to rationalize.

In Table 6, we report results comparing effects for students enrolled in FRL in 2014 (prior to CEP introduction) to those not enrolled. These results are only available for the student fixed effect model. While we acknowledge that this model is based on a distinctive and non-representative sample, we note that the findings are consistent with the idea that CEP reduces suspensions by mitigating the stigma associated with free or reduced-price school meals. Students previously enrolled in FRL experience a large drop in suspension probability in the 9th grade panel, while the effect for those not enrolled is a precise zero. When we interact FRL enrollment with low-income status in Panel C, we see that FRL-enrolled students experience large negative effects even when they are not low-income,[xiii] while low-income students not enrolled in FRL experience smaller (and not statistically significant) effects. The biggest effect of CEP is on those who previously received free or reduced-price lunch, consistent with the stigma-reduction pathway. If CEP reduced suspension by improving nutrition, we would expect benefits to accrue to low-income students not previously receiving lunch supports.

[TABLE 6 AROUND HERE]

Discussion

In this paper, we investigate how CEP, a federal program that allows schools to provide free meals to all students regardless of their family's socioeconomic background, affects participating schools' disciplinary climate. Drawing upon prior research about school meals, the operation of social class in schools, and the links between stigma and exclusionary school discipline, we hypothesize that CEP implementation (1) reduces the incidence of suspensions in participating schools, (2) particularly for students from low-income backgrounds who had previously received free or reduced-price meals and (3) particularly for students from minoritized racial and ethnic groups.

To test these hypotheses, we construct a unique student-by-year-level panel data resource that links more than 2 million annual records of K-12 students enrolled in Oregon public schools between 2009-10 and 2016-17 with rich administrative data records describing their families' household income and social safety net program participation. We use these data to estimate the effects of attending a school that has implemented CEP. Our models use a difference-in-differences design, in which we compare changes in student suspension probabilities in schools that implemented CEP versus those that were eligible to participate in the universal meals program but chose not to.

This estimation strategy hinges on the identifying assumption that, in the absence of CEP implementation, suspension rates in CEP-participating schools would have followed parallel trends to suspension rates in schools that did not participate in CEP. While suspension rates fell rapidly across Oregon during the years leading up to CEP's launch, we find no evidence to suggest that the pace of change diverged between schools that implemented CEP and those that did not in the pre-treatment period.

Consistent with our first hypothesis, we find that suspension probabilities diverge between students enrolled in schools that implement CEP and students in eligible schools that choose not to implement the program. This finding indicates that CEP implementation has a measurable protective effect on suspension rates in participating schools. The CEP effects that we identify are somewhat smaller than the protective effect associated with rigorously evaluated large scale restorative justice implementation efforts (Augustine et al. 2018; Davison et al. 2022). However, we do not see CEP as an alternative to these more targeted programs. Rather, we suspect that universal meal provision – a policy designed to improve student nutrition and reduce administrative burden – can contribute to a broader effort to reform school disciplinary climates by reducing the stigma experienced by economically disadvantaged students.

Consistent with our second hypothesis, we find that CEP implementation has particularly pronounced protective effects on suspensions for students from low-income families, defined as students whose families either participated in SNAP or reported total household income less than 1.85 times poverty in IRS tax filings. These findings suggests that universal meal provision can help narrow persistent educational opportunity gaps associated with family economic disadvantage.

But it is important to note that even in schools with means-tested school meals, many students from low-income families do not enroll for free or reduced-price meals and many students from relatively high-income families do enroll for free or reduced-price meals. Further analyses that disaggregate the effects of CEP on students who had enrolled in free or reduced-price lunch programs in 2013-14, the year prior to the program's availability in Oregon, indicate that the program's protective effects are concentrated among students who had previously received free or reduced-price meals, regardless of their family's household income.

We further find that CEP has more pronounced protective effects for Hispanic students than White students, consistent with our third hypothesis. In summarizing these findings, we note that we are unable to estimate the effect of CEP implementation on Black students with precision. This is a major limitation, since anti-Black racism is a large and deeply embedded source of inequality and substantial attention has been dedicated to narrowing persistent Black/white disciplinary disparities. Our findings cannot speak convincingly to whether or not CEP implementation can contribute to this effort. Future research should investigate whether these effects hold in settings with larger Black student populations.

Viewing universal meal provision as an intervention that mitigates a source of stigma in the school helps to explain how the policy's effects reach beyond student nutrition to promote equity at scale in the disciplinary arena. Traditional school meal programs create categories based on students' reported family income and then enact those categories daily as students claim their meals (Best 2017, Domina et al. 2018, Weaver-Hightower 2011). But these categories are malleable. Universalizing school meals via CEP reduces the distinction between students who receive free or reduced-price meals and other students by collapsing all students into the free meal category. Our analyses indicate that the effects of this adjustment in educational categories ripple through school cultures to the particular benefit of groups of students who we might expect to be particularly at risk for socioeconomic stigma.

Of course, CEP does not just manipulate the social categories that operate in schools; it also makes it easier for all students to eat school meals. Since previous research indicates that lunch stigma is a primary deterrent to school meal consumption – for students who qualify for free meals as well as for students who would have to pay for school meals – CEP's nutritional effects are fundamentally intertwined with its stigma-reducing effects. We find it notable that our

analyses indicate that the program's effects are most pronounced for students who were already eligible to receive nutritional supports via traditional means-tested meals programs. Indeed, we believe that CEP effects on students who had previously enrolled in free or reduced-price lunch despite coming from non-poor families provide the clearest indication in our analyses of the link between meal stigma and school discipline, since we suspect that the nutritional and bureaucratic benefits of universal meal provision may be relatively muted for these relatively advantaged students. We acknowledge, however, that our analyses cannot conclusively separate stigma and nutrition as mechanisms for program effects. We see the examination of these two highly linked program pathways as a particularly promising avenue for future research.

Similarly, our data do not allow us to distinguish CEP's effect on student behavior from its effects on teacher responses to student behavior. After all, both administrative burden and stigma reduction channels apply to teachers. CEP could change teachers' perceptions of students and their behavior, or it could simply free up staff time to monitor behavior in the lunch room. This is another important consideration for future work.

As more schools adopt CEP and concerns about food insecurity mount, political pressure to extend the program and offer free meals to all students is growing. In the spring of 2020, the USDA enacted a temporary waiver that allowed schools to provide free meals to all children, regardless of income eligibility, in order to support students and families during the COVID-19 pandemic. In the pandemic's wake, policymakers at the state and federal level are considering a wide range of policies to further expand access to free school meals. Our findings cannot speak directly to the potential effects of these policies, since CEP's design only allows us to estimate the effects of universal meal provision in schools with relatively high concentrations of poor students. We further note that CEP rolled out in Oregon amidst a broader effort to reform school

disciplinary practices. It is possible that CEP implementation interacted with these disciplinary reform efforts in ways that we cannot model empirically.

Despite these limitations, we believe our findings highlight the potential to create more equitable schools by reducing the salience of labels that evoke stigma.

Appendix A: Variable Construction

Table 1 provides an overview of our data and key variables. Student-level variables include information on our key outcome – suspension – as well as demographic characteristics (from ODE data), school mobility rates (ODE data), SNAP enrollment and benefits (Oregon SNAP data), and family income (IRS data). School characteristics are derived primarily from ODE data, but school median adjusted gross income is measured from IRS records. School use of Provision 2 – which indicates participation in a different NSLP program that facilitated universal breakfast and lunch provision in certain cases prior to CEP – is identified from public records.

While the variables from ODE are standard in K-12 administrative data, the variables derived from SNAP and IRS records require more detailed explanation.

Enrolled in SNAP in March. This indicates students whose households received nonzero SNAP benefits in March of the relevant school year. We use March SNAP receipt, because SNAP enrollments as of April 1 of each year are used to determine school eligibility for CEP, and March receipt was our best proxy for that point-in-time measure.

Total annual family SNAP benefits. This is the total SNAP benefits received by the student's household from August through July (corresponding with the relevant school year).

Adjusted gross income (back-filled). This is the adjusted gross income reported on the tax return (IRS Form 1040) on which the student is claimed as a dependent. Tax records reflect calendar years, so we link to the tax record corresponding to fall of each school year (e.g., student records for the 2014-15 school year are linked to 2014 tax records). Since some households do not file taxes each year, we take the adjusted gross income from the most recent tax record on which the student appears, up to four years prior to the reference year. To ensure comparability, we adjust

the prior year records for inflation using the Consumer Price Index, so that any backfilled income value is reported in reference year dollars.

Low income. This is a composite indicator that leverages both Oregon SNAP and IRS records. Specifically, students are designated as low income if (a) they are enrolled in SNAP at any point during the school year,[xiv] or (b) they are not enrolled in SNAP but their family's income-to-poverty ratio falls below 1.85 times the poverty line. To calculate the family income-to-poverty ratio, we assign a poverty line based on family size proxied from the student's most recent tax return record, and divide the adjusted gross income (back-filled) value by the poverty line.

Appendix B: Estimating equations

In recent years, the econometrics literature has paid substantial attention to the complexity of identifying treatment effects using a DiD model when entry into treatment is staggered across time (Goodman-Bacon 2021). In particular, the canonical model with a single treatment effect is inadequate when entry is staggered and treatment effects potentially build over time.

We have three treatment cohorts – schools that begin CEP in 2014-15 (the 2015 cohort), 2015-16 (the 2016 cohort), or 2016-17 (the 2017 cohort) – and, thus, must address modeling challenges associated with staggered entry. We do so via two different approaches:

First, we take the approach outlined by Wooldridge (2021) which establishes clear control groups and allows treatment effects to vary by cohort and time period of treatment. This approach enables us to look for evidence of treatment effects that build over time, as we hypothesize they might, while still enabling estimation of a single treatment effect where appropriate. To support our assumption that treatment effects do not vary across cohorts (i.e., CEP has the same effect in the first year regardless of which year a school starts the program), we first estimate a disaggregated version of the central model with cohort-by-period treatment effects.

This, which allows us to test whether treatment effects, in fact, vary across cohorts. model takes the following form:

$$y_{itj} = \eta + \sum_{s=11}^{17} \theta_s fs_t + \tau_{15,15} d_{i,15} f15_t + \tau_{15,16} d_{i,15} f16_t + \tau_{15,17} d_{i,15} f17_t$$

$$+\tau_{16,16} d_{i,16} f16_t + \tau_{16,17} d_{i,16} f17_t + \tau_{17,17} d_{i,17} f17_t + \gamma_j + \epsilon_{it}$$

y_{it}, the dependent variable in this model, as in all models we estimate in this paper, is an indicator for whether student i is suspended in year t while enrolled in school j. Each treatment

cohort receives its own treatment indicator, $d_{i,y}$, where y=15 if the school entered treatment in 2014-15 academic year, and so on. In addition, the model includes academic year indicators (fs_t where s ranges from 2011 through 2017, with 2010 excluded as a reference year), and the interaction of treatment indicators with academic year indicators *for academic years in which the cohort is participating in CEP*. The coefficients on these interaction terms ($\tau_{15,15}, \tau_{15,16}, \ldots, \tau_{17,17}$) capture the effects of CEP for each cohort and in each year the cohort is treated.

Since we find little evidence of treatment variation across cohorts, we additionally estimate the following simplified event study model:

$$(1) \quad y_{itj} = \eta + \sum_{s=11}^{17} \theta_s fs_t + \sum_{p=1}^{3} \tau_p g_{j,p} + \gamma_j + \epsilon_{it}$$

The indicator $g_{j,p}$ flags the year of CEP participation for treated schools, where p ranges from 1-3. The corresponding coefficient, τ_p, represents the period-specific treatment effect.[xv] Our first hypothesis holds that the $\tau_p$coefficients are jointly statistically significantly smaller than zero. Our second hypothesis holds that the value of τ_p should become more negative as p increases. Treatment effects are identified under our estimating assumptions because school fixed effects (γ_j) remove level differences in suspensions rates across schools, and academic year indicators fs_t adjust for time trends in suspension rates common to all CEP-eligible schools (note: index s ranges from 2011 through 2017, with 2010 excluded as a reference year).

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Table 1: Population and sample summary

	All Oregon schools		CEP-eligible schools	
	Mean	Std. dev.	Mean	Std. dev.
A. Student characteristics				
Ever suspended	0.06	0.25	0.08	0.26
Suspension days	0.21	1.28	0.24	1.39
Female	0.49	0.50	0.49	0.50
American Indian/Alaskan Native	0.02	0.13	0.02	0.14
Asian/Pacific Islander	0.05	0.21	0.04	0.20
Black	0.02	0.16	0.03	0.17
Hispanic	0.22	0.41	0.26	0.44
White	0.65	0.48	0.60	0.49
Multiracial	0.05	0.22	0.05	0.21
Moves schools between years (nonstructural)	0.09	0.28	0.09	0.29
Moves schools during year	0.05	0.21	0.05	0.22
Language besides English spoken at home	0.22	0.41	0.27	0.44
Enrolled in SNAP in March	0.29	0.46	0.41	0.49
Total annual family SNAP benefits	1,491	2,573	2,093	2,870
Adjusted gross income (backfilled)	71,760	168,900	49,230	67,910
Low income	0.51	0.50	0.63	0.48
B. School characteristics				
Provision 2	0.10	0.30	0.19	0.39
Avg. school enrollment	721	564	634	484
Avg. district enrollment	15,270	15,320	13,190	14,270
Avg. school median AGI	54,290	27,860	37,930	9,237
Avg. proportion grades K-5	0.46	0.50	0.56	0.50
Avg. proportion grades 6-8	0.23	0.42	0.21	0.41
Avg. proportion grades 9-12	0.31	0.46	0.23	0.42
C. Sample description				
No PIK	0.04	0.19	0	
No low income value	0.06	0.24	0	
N(observations) All	4,483,000			
N(students) All	1,013,000			
N(observations) with PIK	4,312,000			
N(students) with PIK	956,000			
N(observations) with PIK & Low Income value	4,219,000		2,104,000	
N(students) with PIK & Low Income value	945,000		567,000	

Note: "Avg." school characteristics are measured at school level but averaged at student level, resulting in a weighted average of school characteristics in each cell. Estimates and counts are rounded according to Census disclosure guidelines. Approved under CBDRB-FY22-CES014-012 and CBDRB-FY23-CES014-045.

Table 2: Trends in covariates for treatment and control schools

	Ever in CEP				Never in CEP			
	2010	2014	Change (2014-	2017	2010	2014	Change (2014-	2017
A. Student characteristics								
Female	0.49	0.49	0.00	0.49	0.49	0.49	0.00	0.49
American Indian/Alaskan Native	0.03	0.03	-0.01	0.02	0.02	0.02	0.00	0.01
Asian/Pacific Islander	0.04	0.04	0.00	0.04	0.04	0.04	0.00	0.04
Black	0.04	0.04	0.00	0.04	0.03	0.02	0.00	0.02
Hispanic	0.30	0.34	0.04	0.34	0.17	0.21	0.04	0.22
White	0.56	0.51	-0.04	0.50	0.71	0.66	-0.05	0.64
Multiracial	0.02	0.05	0.02	0.05	0.03	0.05	0.02	0.06
Moves schools between years (nonstructural)		0.12		0.11		0.10		0.10
Moves schools during year	0.06	0.06	0.00	0.05	0.05	0.05	0.00	0.04
Language besides English spoken at home	0.33	0.34	0.01	0.34	0.19	0.20	0.01	0.22
Enrolled in SNAP in March	0.45	0.49	0.04	0.42	0.33	0.37	0.04	0.31
Total annual family SNAP benefits	2,405	2,539	134	2,070	1,730	1,878	148	1,512
Adjusted gross income (backfilled)	40,510	42,680	2,170	49,410	50,010	52,930	2,920	60,830
Low income	0.69	0.72	0.03	0.67	0.57	0.60	0.03	0.55
B. School characteristics								
Provision 2	0.40	0.49	0.10	0.01	0.11	0.12	0.01	0.14
Avg. school enrollment	522	534	12	539	720	715	-4	713
Avg. district enrollment	14,510	14,970	460	15,260	11,470	11,820	350	12,140
Avg. school median AGI	31,770	32,720	950	37,650	39,580	40,660	1,080	46,480
Number of students	116,000	118,000	2,000	117,000	145,000	147,000	2,000	145,000
Number of schools	300	300	0	300	350	350	0	350

Note: "Avg." variables are measured at school level but averaged at student level, resulting in a weighted average of school characteristics in each cell; counts are rounded according to Census disclosure guidelines. Approved under CBDRB-FY22-CES014-012 and CBDRB-FY23-CES014-045.

Table 3: Suspension probability pre-trends analysis, linear probability model coefficients for academic year indicators interacted with CEP implementation cohort

Academic year	No interaction	2015 cohort * year indicator	2016 cohort * year indicator	2017 cohort * year indicator
2010	0.020***	**0.001**	**0.020**	**-0.013**
	(0.003)	**(0.007)**	**(0.014)**	**(0.010)**
2011	0.020***	**-0.002**	**0.009**	**-0.015**
	(0.004)	**(0.006)**	**(0.011)**	**(0.012)**
2012	0.014***	**0.002**	**0.005**	**-0.005**
	(0.003)	**(0.005)**	**(0.005)**	**(0.016)**
2013	0.007***	**-0.001**	**0.004**	**-0.001**
	(0.002)	**(0.004)**	**(0.003)**	**(0.007)**
2014 (reference)	--	--	--	--
2015	-0.003	-0.004	**0.004**	**-0.003**
	(0.003)	(0.005)	**(0.008)**	**(0.011)**
2016	-0.004	-0.004	-0.005	**-0.000**
	(0.002)	(0.004)	(0.008)	**(0.013)**
2017	0.000	-0.010*	-0.010	-0.005
	(0.002)	(0.004)	(0.009)	(0.015)
Constant	0.069***			
	(0.001)			
School fixed effects	Yes			
Covariates	No			
Observations	2104000			
r2	0.052			

Note: this table displays the full set of coefficients from a single regression. The "No interaction" column displays coefficients on academic year indicators; the "2015 cohort * year indicator" column displays coefficients on an indicator for membership in the 2015 CEP cohort interacted with academic year indicators. The reference (omitted) year is 2014. Cohort indicators (not interacted) are subsumed by school fixed effects. Standard errors are clustered at the school district level and are shown in parentheses. * p<0.05; ** p<0.01; *** p<0.001. Approved under CBDRB-FY23-CES014-045.

Table 4: Linear probability model, Treatment effects on probability of suspension

	(1) Full sample	(2) Full sample	(3) Grades K-5	(4) Grades 6-8	(5) Grades 9-12	(6) Elementary panel	(7) 9th grade panel
First period of treatment	-0.006						
	(0.004)						
Second period of treatment	-0.006						
	(0.003)						
Third period of treatment	-0.010**						
	(0.003)						
Aggregated treatment effect		-0.007*	-0.004	-0.019**	-0.025*	0.001	-0.029**
		(0.003)	(0.002)	(0.007)	(0.011)	(0.002)	(0.010)
Constant	0.077***	0.077***	0.036***	0.146***	0.116***	0.023***	0.064***
	(0.000)	(0.000)	(0.000)	(0.001)	(0.001)	(0.001)	(0.002)
School fixed effects	Yes	Yes	Yes	Yes	Yes	--	--
Student fixed effects	No	No	No	No	No	Yes	Yes
Observations	2104000	2104000	1181000	448000	475000	170000	37000
r2	0.052	0.052	0.013	0.028	0.032	0.418	0.413
P-value, test: effects are equal across periods	0.427	--	--	--	--	--	--
Basline suspension rate	0.073	0.073	0.039	0.155	0.116	0.020	0.088

Note: The baseline suspension rate is the suspension rate of students in the relevant subsample in treated schools in 2014 (prior to the start of CEP). Indicators for academic year are suppressed. School fixed effects are subsumed by student fixed effects in regressions 6 and 7. Standard errors are clustered at the school district level and are shown in parentheses. * p<0.05; ** p<0.01; *** p<0.001. Approved under CBDRB-FY23-CES014-045.

Table 5: Linear predictions, heterogeneous treatment effects

	All	Grades K-5	Grades 6-8	Grades 9-12	Elementary panel	9th grade panel
A. Low income status						
Not low income	-0.002	-0.003	-0.017*	-0.017	0.000	-0.014
	(0.004)	(0.002)	(0.007)	(0.011)	(0.002)	(0.008)
Low income	-0.009**	-0.004	-0.018*	-0.029*	0.000	-0.043***
	(0.003)	(0.003)	(0.008)	(0.012)	(0.003)	(0.013)
B. Lunch eligibility proxy						
Ineligible	-0.002	-0.003	-0.014*	-0.020	0.001	-0.021*
	(0.003)	(0.002)	(0.007)	(0.012)	(0.002)	(0.010)
Reduced-price-lunch eligible	-0.007*	-0.005*	-0.022**	-0.022*	-0.003	-0.035*
	(0.003)	(0.003)	(0.008)	(0.011)	(0.003)	(0.017)
Free-lunch eligible	-0.011**	-0.005	-0.021*	-0.031**	0.0004	-0.041**
	(0.004)	(0.003)	(0.008)	(0.012)	(0.003)	(0.015)
C. Race						
White	-0.003	-0.002	-0.012	-0.021	0.007	-0.021**
	(0.003)	(0.003)	(0.008)	(0.011)	(0.003)	(0.008)
Hispanic	-0.014***	-0.009***	-0.027**	-0.027	-0.002	-0.050*
	(0.004)	(0.003)	(0.008)	(0.016)	(0.003)	(0.020)
Black	-0.002	0.009*	0.027	-0.034*	0.017	-0.062
	(0.006)	(0.004)	(0.032)	(0.017)	(0.011)	(0.035)
API	0.003	-0.009**	-0.021*	0.004	0.013	-
	(0.006)	(0.003)	(0.009)	(0.019)	(0.010)	
AIAN	0.014	0.014	-0.041*	-0.001	0.006	0.038
	(0.022)	(0.010)	(0.018)	(0.031)	(0.011)	(0.040)
Multiracial	0.000	0.004	-0.006	-0.021	0.001	-
	(0.005)	(0.004)	(0.009)	(0.017)	(0.007)	
School fixed effects	Y	Y	Y	Y	-	-
Student fixed effects	N	N	N	N	Y	Y

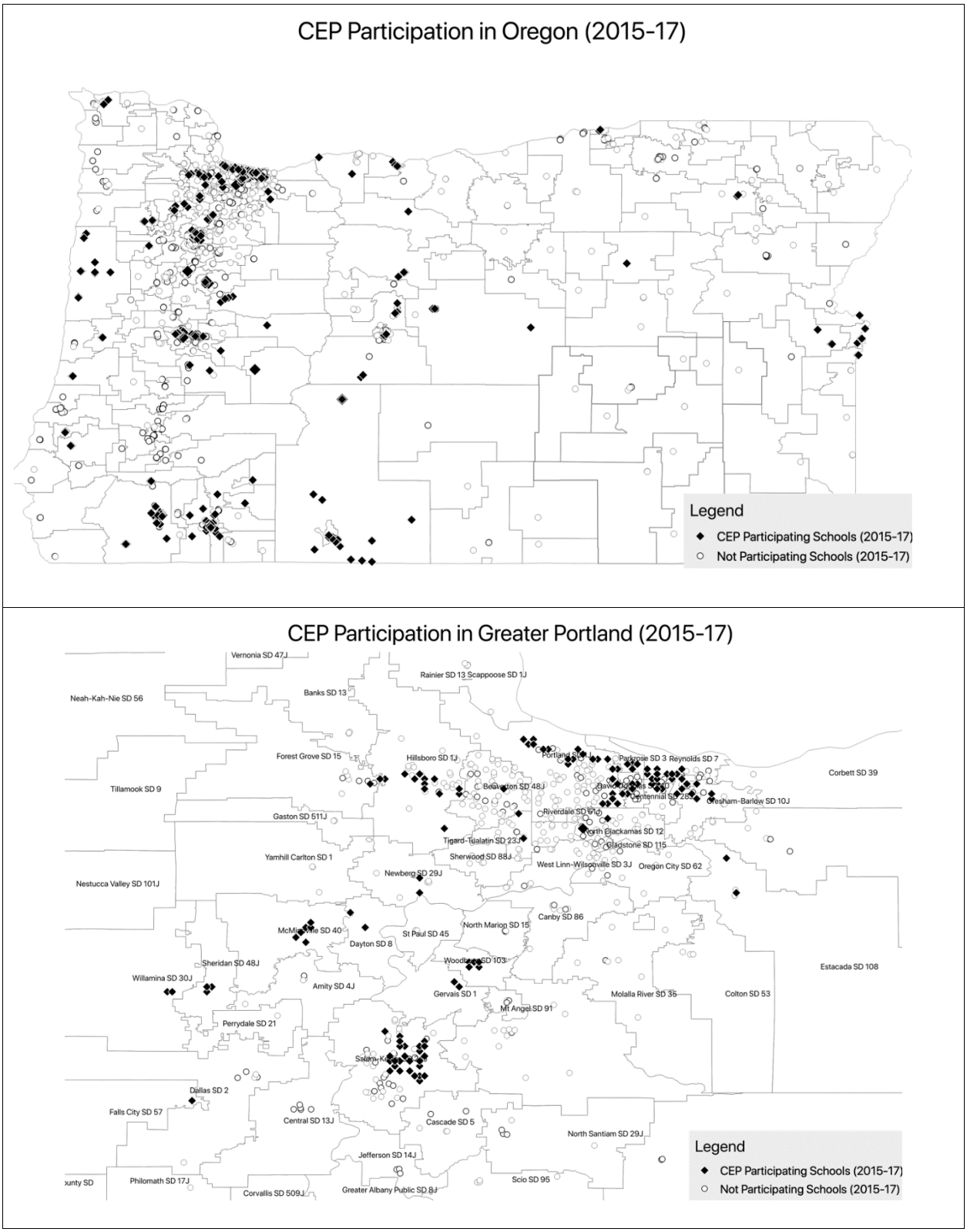
Note: Each panel reports linear predictions from a regression that includes the relevant characteristic interacted with treatment status and academic year indicators, as shown in Model (3). Linear predictions for Asian/Pacific Islander and multiracial students in the 9th grade panel are suppressed for privacy. Standard errors in parentheses; * p<0.05; ** p<0.01; *** p<0.001. Approved under CBDRB-FY23-CES014-045.

Table 6: Linear Predictions, heterogeneous treatment effects based on 2014 free or reduced-price lunch enrollment

	Elementary panel	9th grade panel
A. FRL enrollment status in 2014		
Not enrolled in FRL	0.002	-0.001
	(0.002)	(0.008)
Enrolled in FRL	-0.001	-0.042**
	(0.003)	(0.013)
B. Low income status in 2014		
Not low income	0.002	-0.019*
	(0.002)	(0.008)
Low income	-0.001	-0.036**
	(0.003)	(0.014)
C. FRL/low income contrast		
Not enrolled or low income	0.002	0.001
	(0.002)	(0.008)
Not enrolled but low income	0.001	-0.011
	(0.007)	(0.021)
Enrolled but not low income	0.001	-0.056**
	(0.004)	(0.020)
Enrolled and low income	-0.001	-0.039**
	(0.003)	(0.014)

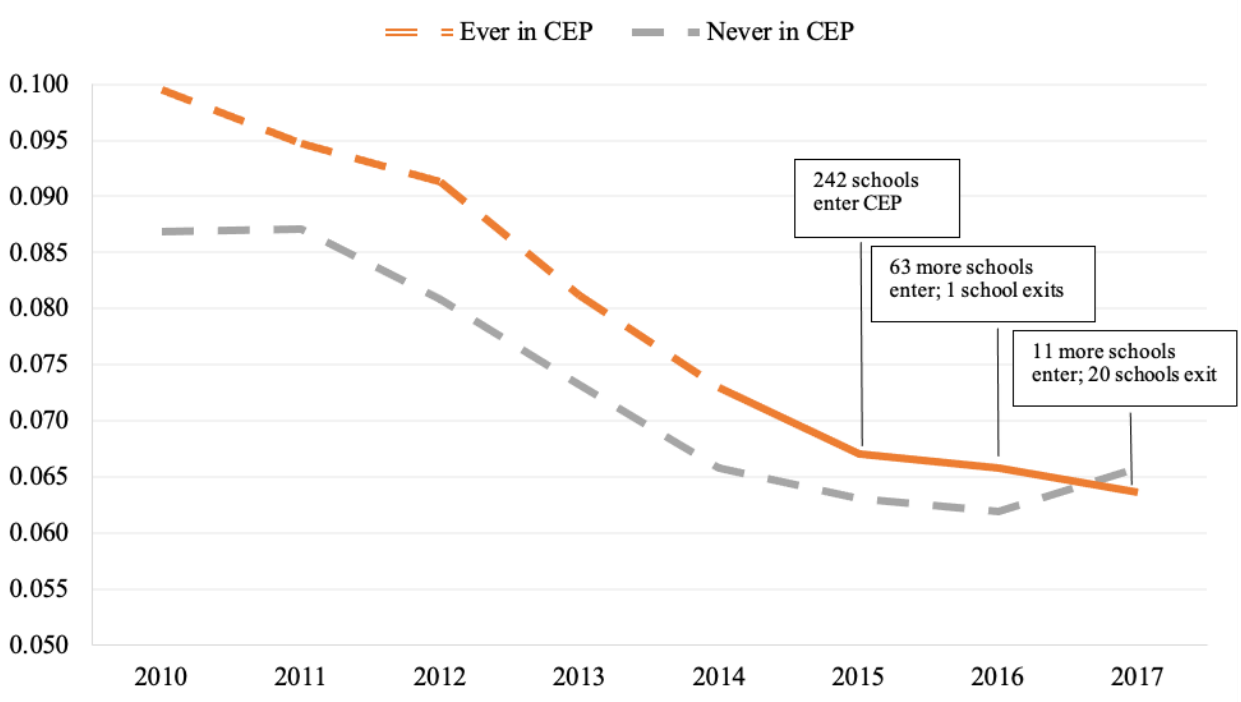
Note: Each panel reports linear predictions from a regression that includes the relevant characteristic interacted with treatment status and academic year indicators, as shown in Model (3). Underlying regressions include student fixed effects. Standard errors in parentheses; * $p<0.05$; ** $p<0.01$; *** $p<0.001$. Approved under CBDRB-FY23-CES014-045.

Figure 1: Map of schools by CEP participation



Source: Oregon Department of Education.

Figure 2: Trends for suspension rates



Note: Approved under CBDRB-FY22-CES014-012

Appendix Table 1: Suspension rate pre-trends analysis, linear probability model coefficients for academic year indicators interacted with CEP implementation cohort

Panel A: Grades K-5

Academic year	No interaction	2015 cohort * year indicator	2016 cohort * year indicator	2017 cohort * year indicator
2010	0.01***	**0.00**	**0.01**	**-0.01**
	(0.00)	**(0.00)**	**(0.00)**	**(0.01)**
2011	0.01***	**0.00**	**0.01***	**-0.01**
	(0.00)	**(0.00)**	**(0.01)**	**(0.01)**
2012	0.01***	**0.00**	**-0.00**	**-0.01**
	(0.00)	**(0.00)**	**(0.00)**	**(0.01)**
2013	0.00	**-0.00**	**0.00**	**-0.01*****
	(0.00)	**(0.00)**	**(0.00)**	**(0.00)**
2014 (reference)	--	--	--	--
2015	0.00	-0.00	**-0.00**	**-0.01**
	(0.00)	(0.00)	**(0.01)**	**(0.01)**
2016	-0.00	-0.00	-0.00	**-0.01*****
	(0.00)	(0.00)	(0.01)	**(0.00)**
2017	0.00	-0.01	-0.01	-0.02***
	(0.00)	(0.00)	(0.01)	(0.00)
Constant	0.03***			
	(0.00)			
School FEs	Yes			
Covariates	No			
Observations	1181000			
r2	0.0132			

Panel B: Grades 6-8

Academic year	No interaction	2015 cohort * year indicator	2016 cohort * year indicator	2017 cohort * year indicator
2010	0.03***	**0.01**	**0.02**	**-0.01**
	(0.01)	**(0.01)**	**(0.02)**	**(0.04)**
2011	0.02***	**0.01**	**0.02**	**-0.01**
	(0.01)	**(0.01)**	**(0.02)**	**(0.03)**
2012	0.02***	**0.01**	**0.01**	**0.01**
	(0.01)	**(0.01)**	**(0.01)**	**(0.06)**
2013	0.01	**0.00**	**0.02**	**0.04**
	(0.00)	**(0.01)**	**(0.01)**	**(0.03)**
2014 (reference)	--	--	--	--
2015	-0.02***	-0.00	**0.02**	**-0.02**
	(0.01)	(0.01)	**(0.01)**	**(0.01)**
2016	-0.01	-0.01	-0.00	**-0.03**
	(0.01)	(0.01)	(0.02)	**(0.02)**
2017	0.00	-0.02*	-0.03	0.00
	(0.01)	(0.01)	(0.02)	(0.03)
Constant	0.14***			
	(0.00)			
School FEs	Yes			
Covariates	No			
Observations	448000			
r2	0.0279			

Panel C: Grades 9-12

Academic year	No interaction	2015 cohort * year indicator	2016 cohort * year indicator	2017 cohort * year indicator
2010	0.03***	**0.01**	**0.10**	--
	(0.01)	**(0.02)**	**(0.06)**	
2011	0.04***	**0.01**	**0.06**	--
	(0.01)	**(0.02)**	**(0.04)**	
2012	0.02**	**0.01**	**0.02**	--
	(0.01)	**(0.03)**	**(0.01)**	
2013	0.01	**0.01**	**-0.00**	--
	(0.01)	**(0.02)**	**(0.01)**	
2014 (reference)	--	--	--	--
2015	0.00	-0.02	**0.00**	--
	(0.01)	(0.01)	**(0.01)**	
2016	-0.00	-0.02	-0.01	--
	(0.01)	(0.01)	(0.01)	
2017	-0.00	-0.02	0.01	--
	(0.01)	(0.01)	(0.02)	
Constant	0.10***			
	(0.00)			
School FEs	Yes			
Covariates	No			
Observations	475000			
r2	0.0331			

Note: Each panel in this table displays the full set of coefficients from a single regression. The "No interaction" column displays coefficients on academic year indicators; the "2015 cohort * year indicator" column displays coefficients on an indicator for membership in the 2015 CEP cohort interacted with academic year indicators. The reference (omitted) year is 2014. Cohort indicators are subsumed by school fixed effects. Standard errors in parentheses. * p<0.05; ** p<0.01; *** p<0.001. 2017 cohort indicators are suppressed in Panel C in compliance with Census disclosure requirements. Approved under CBDRB-FY23-CES014-045

Appendix Table 2: Linear probability model, Disaggregated suspension treatment effects

	All	Grades K-5	Grades 6-8	Grades 9-12	Elementary panel	9th grade panel
A. Cohort-by-treatment-period effects	(1)	(2)	(3)	(4)	(5)	(6)
Cohort 2015 in year 2015	-0.004	-0.004	-0.010	-0.024*	-0.001	-0.041*
	(0.004)	(0.003)	(0.010)	(0.012)	(0.002)	(0.016)
Cohort 2015 in year 2016	-0.004	-0.002	-0.015	-0.022	0.003	-0.022*
	(0.004)	(0.003)	(0.009)	(0.013)	(0.003)	(0.010)
Cohort 2015 in year 2017	-0.010***	-0.008**	-0.030***	-0.025*	-0.002	-0.035*
	(0.003)	(0.003)	(0.008)	(0.012)	(0.004)	(0.016)
Cohort 2016 in year 2016	-0.012	-0.000	-0.017	-0.039	0.009	-0.014
	(0.000)	(0.006)	(0.016)	(0.024)	(0.005)	(0.015)
Cohort 2016 in year 2017	-0.016	-0.002	-0.042*	-0.022	-0.001	-0.031*
	(0.009)	(0.007)	(0.019)	(0.028)	(0.007)	(0.015)
Cohort 2017 in year 2017	0.000	-0.012**	0.006	0.079	0.010*	-
	(0.010)	(0.004)	(0.033)	(0.065)	(0.005)	
Constant	0.077***	0.036***	0.146***	0.116***	0.023***	0.065***
	(0.000)	(0.000)	(0.001)	(0.001)	(0.001)	(0.002)
School fixed effects	Yes	Yes	Yes	Yes	--	--
Student fixed effects	No	No	No	No	Yes	Yes
Observations	2104000	1181000	448000	475000	170000	37000
r2	0.052	0.013	0.028	0.032	0.418	0.413
P-value, test: first period effects are equal	0.641	0.169	0.815	0.253	0.107	--
P-value, test: second period effects are equal	0.213	0.982	0.173	0.991	0.579	0.589
P-value, test: effects for 2015 cohort are constant	0.048	0.052	0.059	0.939	0.055	0.162
P-value, test: effects for 2016 cohort are constant	0.468	0.608	0.168	0.067	0.325	0.021
B. Treatment-period effects	(7)	(8)	(9)	(10)	(11)	(12)
First period of treatment	-0.006	-0.004	-0.012	-0.027*	0.000	-0.032**
	(0.004)	(0.003)	(0.008)	(0.013)	(0.002)	(0.012)
Second period of treatment	-0.006	-0.002	-0.022*	-0.021	0.002	-0.026**
	(0.003)	(0.003)	(0.010)	(0.012)	(0.003)	(0.009)
Third period of treatment	-0.010**	-0.007*	-0.029***	-0.026*	-0.002	-0.032*
	(0.003)	(0.003)	(0.008)	(0.012)	(0.004)	(0.015)
Constant	0.077***	0.036***	0.147***	0.116***	0.023***	0.064***
	(0.000)	(0.000)	(0.001)	(0.001)	(0.001)	(0.002)
School fixed effects	Yes	Yes	Yes	Yes	--	--
Student fixed effects	No	No	No	No	Yes	Yes
Observations	2104000	1181000	448000	475000	170000	37000
r2	0.052	0.013	0.028	0.032	0.418	0.413
P-value, test: effects are equal across periods	0.427	0.139	0.194	0.700	0.253	0.665

Note: Indicators for academic year are suppressed. School fixed effects are subsumed by student fixed effects in regressions 5, 6, 11, and 12. Standard errors are clustered at the school district level and are shown in parentheses. * p<0.05; ** p<0.01; *** p<0.001. Cohort 2017 in year 2017 coefficient suppressed in regression 6 in compliance with Census disclosure requirements. Approved under CBDRB-FY23-CES014-045.

[i] Under CEP schools and districts are reimbursed under the "free rate" for 1.6 times their identified student percentage (ISP). Thus, schools with an ISP above 62.5% are fully reimbursed for 100% of their served meals while schools with ISPs at 40% are reimbursed at the "free rate" for 64% of their served meals and at the "paid rate" for the remaining 36% of meals served (USDA 2023). While CEP relies on annual ISP calculations implemented mainly through direct certification data rather than manual recertification of student applications every four years (as required under Provisions 2 and 3), schools closer to the 40% ISP eligibility cutoff are reimbursed less generously (CRS 2020).

[ii] The ODE data report free or reduced-price lunch enrollment via a student-by-year level "economic disadvantage" indicator. In the years prior to CEP implementation, this measure largely reflects student free or reduced-price lunch enrollment. However, after the introduction of CEP, this measure's properties vary substantially across schools.

[iii] This is a probabilistic matching approaching conducted by the Census Bureau that leverages personally identifiable information in the raw administrative data to assign an anonymous unique identifier known as a PIK. After the assignment of PIKs, personally identifiable information is removed to protect confidentiality and these anonymized files can be accessed by approved researchers. For more information on the Person Identification Validation System, see Wagner and Layne (2014).

[iv] We use logistic regression to estimate the probability of having a PIK as a function of gender, race, grade, age, and whether the student's native language is English. We run this model separately in each academic year. Then we construct weights for the sample with PIKs that are the inverse of the predicted probability of having a PIK for that student-year. Assuming the covariates above capture heterogeneity in PIK receipt, these weights allow us to generalize estimated treatment effects to all students, not just those with PIKs. Thus, we include them in all of our treatment effect estimates. Ultimately, since the underlying PIK rate is quite high, these weights have little effect on our estimated treatment effects.

[v] In ongoing research, authors find that students with PIKs who appear in neither IRS nor SNAP data exhibit similar characteristics to the average student. Thus, an assumption that these students are universally low-income is likely worse than an assumption of missing-at-random.

[vi] Index s ranges from 2011 through 2017, with 2010 excluded as a reference year for notational simplicity, though, in practice, any pre-treatment year can be designated reference year, and, in our parallel trends test below, we opt to omit 2014.

[vii] These indicators are nested in the $Treat_{ij}$ variable in Model (1): summing across the three period indicators or the six cohort-by-year indicators produces that single treatment indicator.

[viii] Selective exit of schools from the CEP program introduces an additional threat to estimating the effect of CEP via a DiD design. When schools enroll in CEP, they remain enrolled for four years by default. Schools can, however, opt out of the program earlier. This may occur if, for example, the costs associated with participating is CEP are greater than schools anticipated when enrolling in the program, creating a funding gap. Since exit occurs between school years, there is no exit during the first treatment period. Just one school exits from the 2015 cohort in the 2015-16 school year, while 20 (of approximately 300) schools exit the 2015 and 2016 cohorts in the 2016-17 school year. We opt to ignore this exit and, thus, treat exiting schools as though they remained in CEP. This potentially introduces a conservative bias to our estimates since there is no theoretical reason to think that offering universal free lunch one year and then removing it the next leads to improved suspension outcomes relative to offering universal free lunch in both years. If schools stop providing universal free lunch, they likely lose both potential benefits of CEP: improved food access and stigma reduction. It remains possible, however, that schools find ways to maintain aspects of CEP they found beneficial as they transition back to the old reimbursement system.

[ix] We note that the inclusion of these subgroups changes the parallel trends assumption: For the results of Model (3) to yield unbiased estimates of the effect of CEP adoption by subgroup, we must assume that *subgroup* outcomes across treatment/control groups would move in parallel if not for CEP.

[x] We note, however, that the enrollment patterns in Provision 2 diverge between treatment and control schools. Nearly half of treatment school students in 2014 had exposure to some level of universal free breakfast or lunch supports via Provision 2 prior to the implementation of CEP, while just 13 percent of students in control schools did. As we noted previously, Provision 2 is the antecedent to CEP, comes with less financial support, and predominantly offers free breakfast, not lunch. It makes sense that schools using Provision 2 would be more inclined to join CEP. The enormous drop off in Provision 2 participation among treatment schools by 2017 is mechanical – CEP schools must exit Provision 2 – while control schools saw slight growth in Provision 2 use, suggesting it remains an attractive alternative to CEP in certain cases.

[xi] The coefficients on the cohort interactions in years prior to treatment are particularly close to zero for the 2015 and 2017 cohorts, offering strong support of the parallel trends assumption for these cohorts. The coefficients for the 2016 cohort are not statistically significant, but are positive in magnitude, suggesting a slower pre-treatment decline in suspension for schools in the 2016 cohort compared to control schools. 77 percent of treatment schools are in the 2015 cohort.

[xii] See Appendix A for the specific definition of this variable.

[xiii] In this context, "not low income" means that the student was not enrolled in SNAP and, in the tax records observed for their family, earnings exceeded 1.85 times poverty.

[xiv] Typical SNAP guidelines require households have income less than 1.3 times the federal poverty line, but here, since we include households that may have only been SNAP-eligible for part of the year, we designate as low-income some households with income above 1.3 times the poverty line, and, potentially, above 1.85 times the poverty line. However, our analyses suggest that the vast majority of these families have annual income below 1.85 times poverty.

[xv] Since our data ends in the 2016-17 school year, not all cohorts are reflected in each treatment period; while all three cohorts contribute to τ_1, the 2015 and 2016 cohorts contribute to τ_2, and only the 2015 cohort contributes to τ_1.